

ID3 Decision Tree Learning for a Robot

“Ask-for-Help” Decision

Jacob A. Geldbach
College of Engineering: CS
University of Idaho
Moscow Idaho, United States
geld8788@vandals.uidaho.edu

Abstract—Placeholder abstract text.

Index Terms—Decision Tree, ID3, Information Gain, Entropy, Supervised Learning, Classification

I. INTRODUCTION

Placeholder introduction text.

II. ALGORITHM DESCRIPTION

Measuring Information

Placeholder text on entropy and information gain.

Handling Undefined Cases

Placeholder text on undefined/edge cases.

Measuring Performance

Placeholder text on the performance-measurement methodology.

III. RESULTS

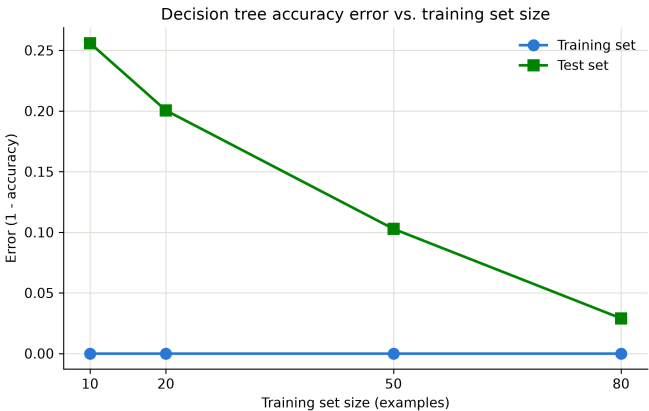


Fig. 1. Accuracy error (1–accuracy) vs. training set size, training-set curve vs. held-out test-set curve (mean over 50 trials per size).



Fig. 2. Recall error (1–recall) vs. training set size, training-set curve vs. held-out test-set curve (mean over 50 trials per size).

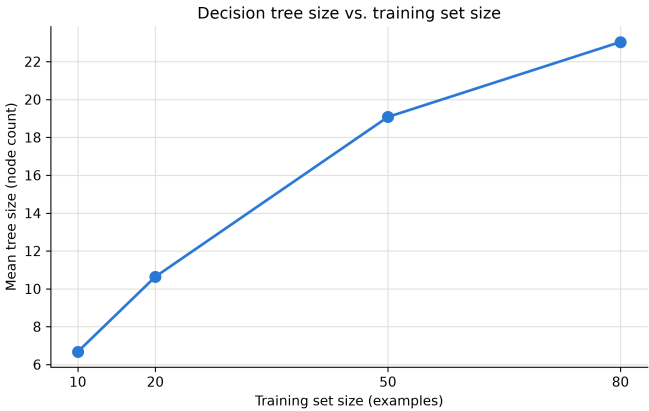


Fig. 3. Mean decision tree size (node count) vs. training set size (mean over 50 trials per size).

IV. DISCUSSION / CONCLUSIONS

Strengths and Weaknesses

Placeholder text on strengths and weaknesses.

Future Improvements

Placeholder text on possible improvements.

TABLE I
ID3 DECISION TREE — SWEEP RESULTS (50 TRIALS PER TRAINING-SET SIZE)

Train Size	Trials	Train Acc.	Test Acc.	Test Prec.	Test Recall	Mean Tree Size	Mean Tree Depth	Most Common Root Attr.
10	50	1.000	0.744	0.800	0.791	6.68	2.62	Safe Situation
20	50	1.000	0.800	0.841	0.832	10.64	3.62	Safe Situation
50	50	1.000	0.897	0.911	0.919	19.08	4.90	Safe Situation
80	50	1.000	0.971	0.972	0.980	23.04	5.08	Safe Situation

USE OF AI

Coding

Placeholder text describing AI use in coding.

Writeup

Placeholder text describing AI use in the writeup.